Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Team 19: Radar Rangers
Bi-Weekly Update 3

Eva Framjee, Neil Garcia, Sterling Light
Sponsor: Dr. Arya Menon
TA: Roman Venegas

Project Summary

- Devices can't detect breathing and heart rate without wearable sensors, causing discomfort.
- Provide a fully contactless method for tracking breathing patterns and heart rate which can be read easily via mobile devices.

Human has irregular breathing pattern or heart rate → Millimeter Wave Radar detects data, microcontroller checks if its normal → If not normal, the data is sent to the mobile application → Mobile Application notifies user or irregularities

Dwight Look College of ENGINEERING TEXAS A&M UNIVERSITY

Sterling's Subsystem

Neil's Subsystem

Eva's Subsystem

Project/Subsystem Overview

PCB Power Supply: Converts 3S 18650 cell supply to 5.15V/8A supply using TI TPS513885VABR; provides power to Radar and Raspberry Pi

Radome: Protects against debris while being electromagnetically transparent at 59 GHz; wall thickness of 1.75mm, made of Teflon (loss tangent < 0.0002, ε_r=2)

Radar Module: TI IWRL6432Boost Uses Chirp Frequency to scan room at 57-61 GHz. Converts reflected analog signal into digital signal and filters out noise based on configuration.

Raspberry Pi: Receives the data from the radar module, parse the data, and displays the desired outputs onto a table, and saves the table locally. Houses AI model that will sift through the desired data, and determine normal behavior or an emergency.

Mobile Application: Programmed in Visual Studio using dart

Stores data into phone (Ad-Hoc) in .csv format

Live graph: readable data of heart and respiratory rate

Stores data back into phone

Past graph: readable data of heart and respiratory rate of up to four weeks

Project Timeline

New PCB design completed, ordered and received 2/16	Full system is soldered before blitz, appropriate battery will be ordered (completed 2/19)	Integration of Radar and Mobile App via BLE (completed 2/16)	Working to have Radar Program run completely through phone (to complete by 2/22)	Final Integration (to complete by 2/19)	Validation (to complete by 3/15)	Demo and Report (to complete by 4/15)

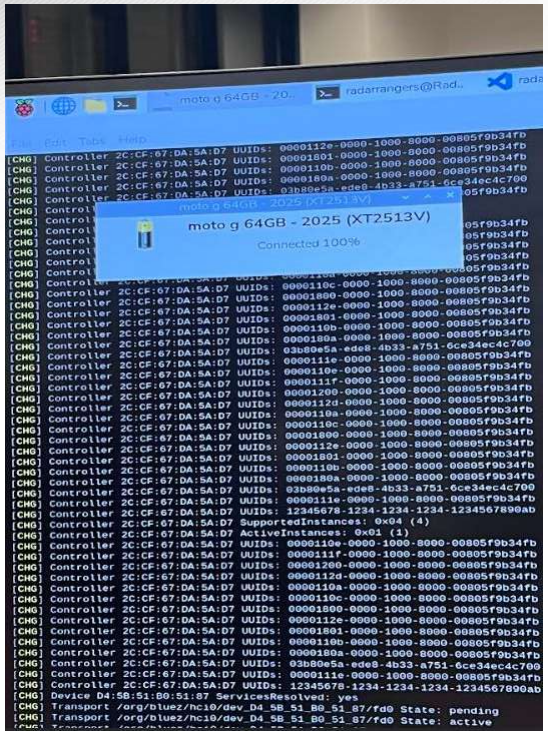
Raspberry Pi and Radar Module

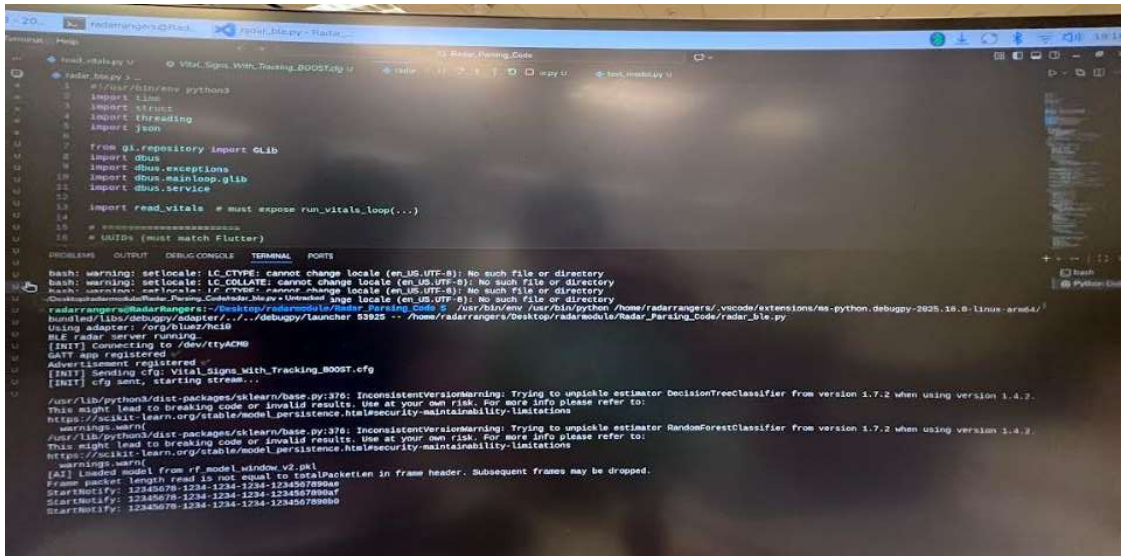
Neil Garcia

Accomplishments since last update 18 hrs of effort	Ongoing progress/problems and plans until the next presentation
- Collaborated with Eva the past two weeks - Created a code that will go through the data picked up by the radar and organize it so it will send to mobile application in organized fashion - Designed a code that will appropriately send data to Mobile application via BLE	- Work with eva to get csv logging on cloud through phone. - Working to have the radar program run completely through the phone rather than running through computer and monitor

Raspberry Pi and Radar Module

Neil Garcia





Mobile Application Design

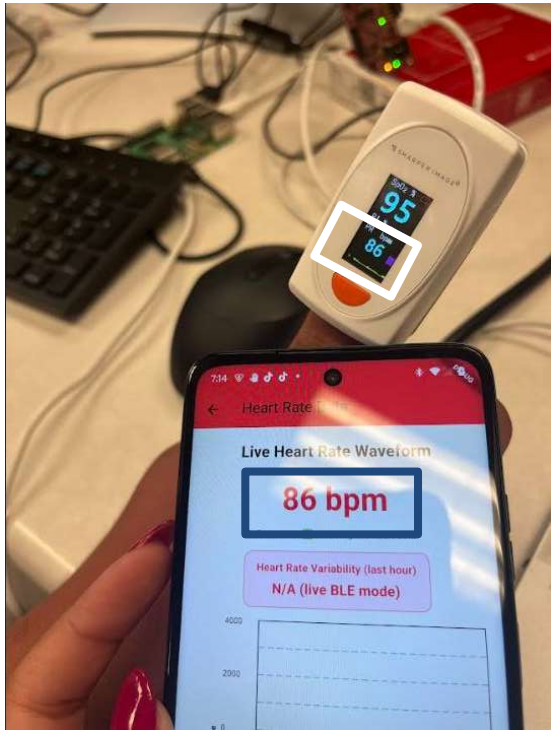
Eva Framjee

Accomplishments since last update 18 hrs of effort	Ongoing progress/problems and plans until the next presentation
- Altered my code to take the data from the raspberry pi and display on correct pages - Collaborated with Neil to ensure our codes match to where correct data is sent and received	- Collaborate with Neil to create a way for the program to run when the app is opened rather than through computers - Fix the problem with live graphs - Make an app logo

Mobile Application Design

Eva Framjee

```
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
D/[FBP-Android](24935): [FBP] onCharacteristicChanged:
D/[FBP-Android](24935): [FBP]   chr: 12345678-1234-1234-1234-1234567890b0
```



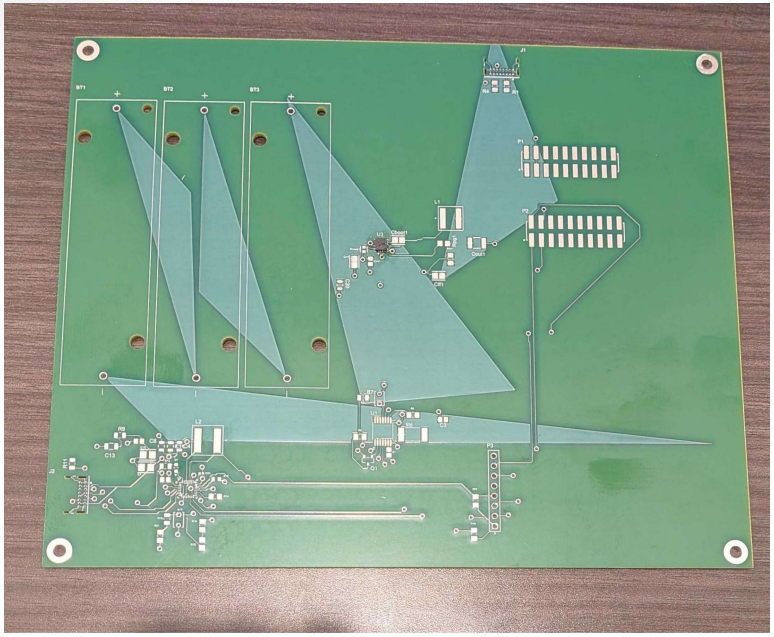
PCB and Radome

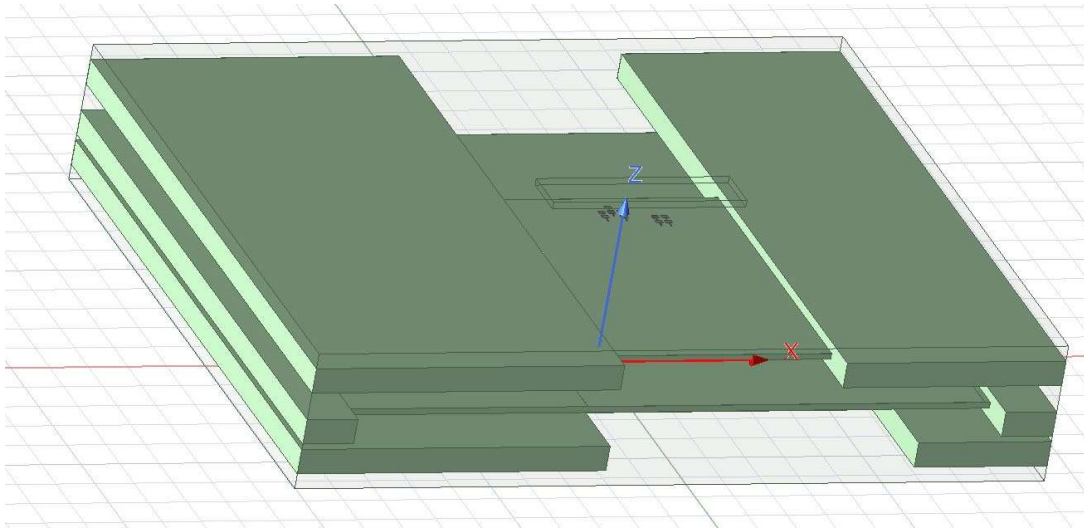
Sterling Light

Accomplishments since last update 31 hrs of effort	Ongoing progress/problems and plans until the next presentation
- Completed new PCB board design - Ordered new PCB board and parts; delivered on Tuesday - Updated radome design to accommodate new PCB	- Complete soldering of power delivery system before blitz - Integrate with rest of the system during blitz - Solder rest of the PCB (battery recharge solution) between blitz and the next presentation - Fabricate radome

PCB and Radome

Sterling Light





Execution & Validation Plan

FR ID	Functional Requirement	Success Criteria	Validation Method	Tested?
3.2.1.4.	App and Radar successfully connected via Wi-Fi	Data from radar is read successfully from the app	Test subject sits in room and monitors via app	YES
3.2.3.7.3	Radar stores data in csv file in cloud that sends data to application	Radar data is successfully stored in a cloud CSV file and fully received by the application without data loss	Run a radar capture and verify the CSV file is created in the cloud and correctly parsed by the app	NO
3.2.1.1.	Detect Heart Rate within 10 beats per minute	Detected heart rate is within +/- 10 BPM of a reference measurement for at least 95% of tests	Compare radar heart rate to a reference sensor across multiple trials and calculate error	NO
3.2.1.1.	Detects Respiratory Rate within 5 breaths per minute	Detected respiratory rate is within +/- 5 breaths per minute of a reference measurement for at least 95% of tests	Compare respiratory rate to a reference breathing measurement during controlled breathing tests	NO
3.2.1.1.	Machine learning model needs to be accurate within 0% false negatives and <1% false positives	The ML model produces 0% false negatives and less than 1% false positives on validation data	Evaluate model performance using labeled test data and compute false negative and false positive rates	NO
3.2.1.3.	Detects person at least 10 ft away	System detects a person at distances up to at least 10 ft with reliable performance	Place a subject at varying distances up to 10 ft and verify detection consistency	NO
3.2.3.6.	PCB effectively powers Raspberry Pi 5	PCB supplies stable power to the Raspberry Pi 5 without resets or undervoltage events	Power the Raspberry Pi 5 from the PCB and monitor operation and voltage stability	NO
3.2.1.4.	App will properly identify and display different incoming data types	App correctly identifies and displays all supported incoming data types without errors	Send test packets for each data type and verify correct display in the app	NO
3.2.3.7.1.	Radome design allows radio waves to propagate with minimal disturbance	Radome causes minimal signal loss, with insertion loss than or equal to 3 dB	Measure or simulate signal transmission through the radome and compare to no-radome baseline	NO
3.2.2.2.4	Radome protects the IWRL6432BOOST Raspberry Pi 5, and PCB Power Supply from damage	Radome fully encloses and protects internal components from physical damage during normal handling	Inspect enclosure fit and perform basic handling tests to confirm continued system operation.	NO

Project Timeline

Task Title	Start Date	End Date	SU1	SU2	SU3	SU4	SU5
Status Update 1: 01/21/26							
Create appropriate axis titles in the app as well as a users manual	01/14/26	02/02/26					
Establish a Wi-Fi Direct system on the Raspberry Pi	01/14/26	02/02/26					
Ensure PCB sustainably powers Pi and radar module	01/14/26	02/02/26					
Status Update 2: 02/04/26							
Setup bluetooth between the Raspberry Pi and the application	02/04/26	02/16/26					
Develop a program that will transfer live data values to phone via Bluetooth	02/04/26	02/16/26					
Create a way to hold all of the system's boards in place	02/04/26	02/16/26					
Status Update 3: 02/18/26							
Test between different people	02/18/26	03/02/26					
Coordinate with other members to connect subsystems together	02/18/26	03/02/26					
Fabricate radome	02/18/26	03/02/26					
Status Update 4: 03/04/26							
Ensure the app successfully connects between the application and the machine	03/04/26	03/23/26					
Fully construct radar apparatus	03/04/26	03/23/26					
Status Update 5: 03/25/26							
Prepare for Demo	03/25/26	04/06/26					
Final Presentation: 04/08/26							
Final Demo							
Showcase							
Final Report							

Thank you and Gig Em!